Average 71/100

Modelling and Control of Multi-body Mechanical Systems

Coursework: Modelling and Control of Wheel Shimmy

Seunghyun Lee (CID: 00843738)

Introduction

This coursework involves modelling and control of wheel shimmy, which is usually implemented in grocery trolley, and aircraft landing gear. The simple model used to demonstrate wheel shimmy, that was simulated in this coursework, is shown in the following figure.

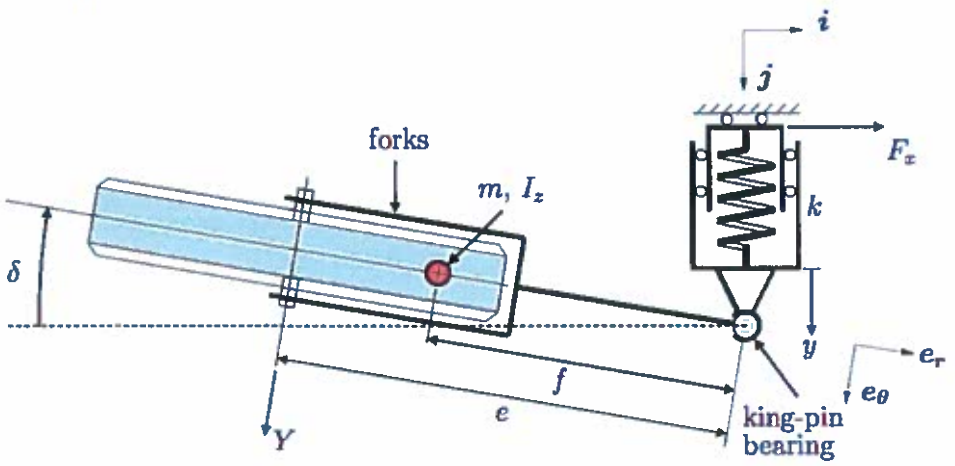


Figure 1. Plan (top) view of a simple system capable of shimmy

Analysing the System

To analyse a certain system, it is good to see the equational relationship among all the variables that represent each components of the system.

Equations of Motion

Question 1 asks for the equations of motion assuming that the tyre can sideslip and that the tyre force is lagged. Equations of Motion derives the physical behaviour of the system by defining the system in function of time.

There are three degree of freedoms, which affects the behaviour of the system, for this mass; therefore, the question expects for three equations of motions.

Equations of motion on the centre of mass is derived from the equation of

$$\sum \boldsymbol{F} = m\ddot{\boldsymbol{r}},$$

where $\boldsymbol{F}$ is the force acting on the centre of mass and $\boldsymbol{r}$ is the direction vector to the centre of mass.

To first compute the right-hand side of the equation, the acceleration vector should be computed. The velocity vector is $\dot{\boldsymbol{r}} = \dot{x}\boldsymbol{i} - f\dot{\theta}\boldsymbol{e_\theta}$.[1] Therefore, the acceleration vector is

$$\begin{aligned}\ddot{\boldsymbol{r}} &= \ddot{x}\boldsymbol{i} - f\ddot{\theta}\boldsymbol{e_\theta} + f\dot{\theta}^2\boldsymbol{e_r} \\ &= \left(\ddot{x} + f\ddot{\theta}\sin\theta + f\dot{\theta}^2\cos\theta\right)\boldsymbol{i} + \left(-f\ddot{\theta}\cos\theta + f\dot{\theta}^2\sin\theta\right)\boldsymbol{j}\end{aligned}$$

through direct differentiation. This computation simply gives,

$$\begin{aligned}m\ddot{\boldsymbol{r}} &= m(\ddot{x}\boldsymbol{i} - f\ddot{\theta}\boldsymbol{e_\theta} + f\dot{\theta}^2\boldsymbol{e_r}) \\ &= m\left(\left(\ddot{x} + f\ddot{\theta}\sin\theta + f\dot{\theta}^2\cos\theta\right)\boldsymbol{i} + \left(-f\ddot{\theta}\cos\theta + f\dot{\theta}^2\sin\theta\right)\boldsymbol{j}\right)\end{aligned}$$

Summary of Comments on Microsoft Word - Modelling and Control of Wheel Shimmy.docx

Page: 2

Number: 1 Author: Simos Evangelou Date: Indeterminate

Simos Evangelou: What about y velocity (y_dot * j)? Also, I suppose you used theta for delta.

on the right-hand side of the equation.

The left-hand side of the equation consists of forces acting on the centre of mass.

$$F = (F_x - Y\sin\theta)\boldsymbol{i} + (F_y - Y\cos\theta - ky)\boldsymbol{j},$$

where

Next, equating both sides of the equation gives us,

$$F_x - Y\sin\theta = m(\ddot{x} + f\ddot{\theta}\sin\theta + f\dot{\theta}^2\cos\theta)$$
$$F_y + Y\cos\theta - ky = m(-f\ddot{\theta}\cos\theta + f\dot{\theta}^2\sin\theta)$$

which are the two equations of motion.

The third equation of motion comes from the motion about the centre of mass. This is given by the equation,

$$\frac{d\boldsymbol{H}}{dt} = \boldsymbol{N}$$

This equation can be expanded into,

$$I_{zz}\ddot{\theta}\boldsymbol{k} = f\boldsymbol{e}_r \times (F_x\boldsymbol{i} + F_y\boldsymbol{j}) - (e-f)\boldsymbol{e}_r \times Y\boldsymbol{e}_\theta$$

Then by substituting this equation with the two previous equations of motion, it gives,

$$\begin{aligned} I_{zz}\ddot{\theta} &= -f\sin\theta\left(m(\ddot{x} + f\ddot{\theta}\sin\theta + f\dot{\theta}^2\cos\theta) + Y\sin\theta\right) + \\ &\quad f\cos\theta\left(m(-f\ddot{\theta}\cos\theta + f\dot{\theta}^2\sin\theta) - Y\cos\theta + ky\right) - (e-f)Y \\ &= -m\ddot{x}f\sin\theta - mf^2\ddot{\theta} + kyf\cos\theta - eY \end{aligned}$$

Putting variables on the same sides now gives,

$$(I_{zz} + mf^2)\ddot{\theta} + m\ddot{x}f\sin\theta + kyf\cos\theta = -eY,$$

which is the last equation of motion.

F_x for the forward speed of the king-pin $\dot{x}$ to have a constant value v

Question 2 required to compute the applied force F_x to get the constant value v in the forward speed of the king-pin $\dot{x}$.

This could be simply derived from one of the equations of motion. Having a constant value of $\dot{x}$, the forward acceleration of the king-pin $\ddot{x}$ has to be 0. Therefore, the first derived equation of motion,

$$F_x - Y\sin\theta = m(\ddot{x} + f\ddot{\theta}\sin\theta + f\dot{\theta}^2\cos\theta),$$

if considered to have this constraint, should become

$$F_x - Y\sin\theta = m(f\ddot{\theta}\sin\theta + f\dot{\theta}^2\cos\theta).$$

Therefore, rearranging the above equation,

$$F_x = m(f\ddot{\theta}\sin\theta + f\dot{\theta}^2\cos\theta) + Y\sin\theta.$$

Equations of Motion (king-pin forward speed constrained and tyre non-holonomic)

Question 3 asks for the equations of motion with king-pin forward speed $\dot{x}$ constrained and the tyre is subject to a non-holonomic constraint.

When the forward speed of the king-pin $\dot{x}$ is constrained to a constant, the acceleration is zero. Therefore, it reduces the system to one degree of freedom, meaning we only have the motion about the centre of mass.

Page: 3

Number: 1 Author: Simos Evangelou Date: Indeterminate
Simos Evangelou: What is Fy?

Number: 2 Author: Simos Evangelou Date: Indeterminate
Simos Evangelou: What about the equation with respect to the tyre lateral force, Y?

Number: 3 Author: Simos Evangelou Date: Indeterminate
Simos Evangelou: How many degrees of freedom do you have before considering any of the constraints here (previously you mentioned three degrees of freedom)? After the forward speed constraint is imposed how many are left?

When the tyre is subject to a non-holonomic constraint, quoting from the coursework booklet, "the lateral velocity of the tyre is constrained to be zero". This means the lateral force response of the tyre Y is going to be zero with the non-holonomic constraint.

Therefore, with these two constraints, we will have first have the equation of motion about the centre of mass.

$$(I_{zz} + mf^2)\ddot{\theta} + m\ddot{x}f\sin\theta + kyf\cos\theta = -eY$$

Then have $\ddot{x}$ and Y to be zero, which will cancel away $m\ddot{x}f\sin\theta$ and $-eY$, leaving the equation of motion of

$$(I_{zz} + mf^2)\ddot{\theta} + kyf\cos\theta = 0$$

Simulation of the System

Building the System

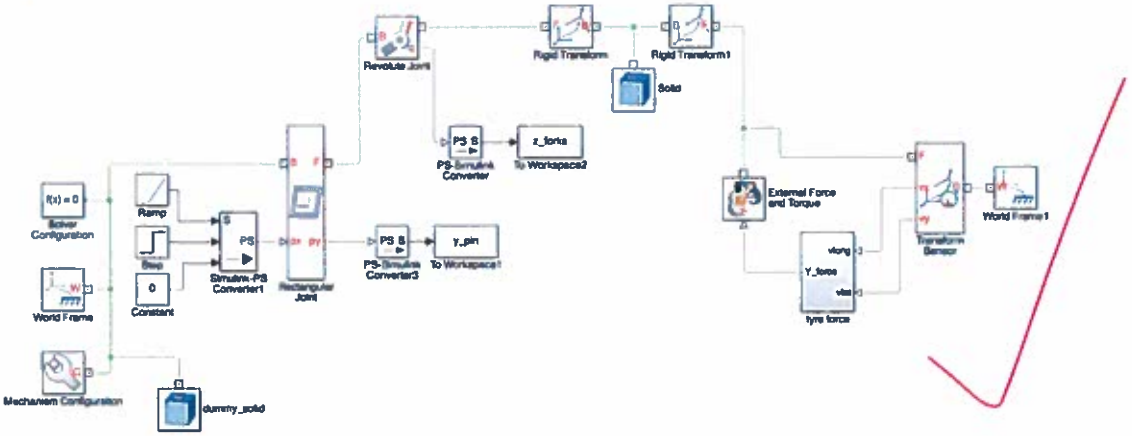


Figure 2. SimMechanics model of Wheel Shimmy

The task in Question 4 involved understanding the components used in SimMechanics models. To connect the separated groups of components from the template from **`coursework_model_template.slx`** to a working model, each component in the template had to be understood and connected correspondingly to the plan view in Figure 1.

Starting from left to right, the *World Frame* in the model represents the ground, which is represented as shaded rectangle on the top right corner of the plan view. Then, the *Rectangular Joint* that is connected to the *World Frame* is a component with two translational freedoms. This has a correspondence to the rectangular body with the spring (vertical translational freedom), and can have whole body moving to F_x direction (horizontal translational freedom). Getting along with that vein, *Revolute Joint* resembles the king-pin bearing, *Rigid Transforms* model mechanical trail e and mass offset f, the *Solid* is the wheel, and *External Force and Torque* with *Transform Sensor* gives the Y torque.

Therefore, connecting each other as corresponding to Figure 1 led to the final system design as Figure 2, which could be compiled correctly.

Number: 1 Author: Simos Evangelou Date: Indeterminate

Simos Evangelou: Not at all. The lateral force Y is now a force of constraint that maintains the nonholonomic constraint.

Examining the Model

Question 5, Examining the SimMechanics model, required the inspection of variable values set for each component. This was able to be done simply with double clicking of the model and looking through the set-up page.

The default value for e and f is related to the length of the mechanical trail and mass offset, which are represented by *Rigid Transforms*. Therefore, double clicking two *Rigid Transforms* showed the set-up with translation of $[0.7\ 0]\ m$ and $[-0.3\ 0]\ m$. Considering that it is the wheel between the two *Rigid Transforms*, and that the 0.7 m one is the one connected to the king-pin bearing, it could be said that $0.7\ m$ is the length of the mechanical trail e, and the $-0.3\ m$ is the length difference between e and the mass offset f.

Therefore, the mechanical trail e has the length of $0.7\ m$. And the mass offset f has the length of $(0.7 + 0.3)\ m$, which gives the total of $1.0\ m$.

The default value for m is the mass of the wheel, which could be determined through the *Solid*. The default value for m was $200\ kg$. The moments of inertia on the z direction I_z was also able to be determined through the *Solid*. The default value for I_z was $150\ kg \cdot m^2$

[1]

Complex Plane Eigenvalues of System with k Variations

Eigenvalues of the linearised state-space model of a system gives the poles of the system which is strongly related to the system's stability. If the eigenvalue takes a negative real number, the system is stable, marginally stable if the real equals to zero, and unstable if the real is larger than zero. And for imaginary axis, the larger the imaginary part of the eigenvalue the larger the frequency of the oscillation the system will take. This means, for a stable eigenvalue, the larger the imaginary part the faster the system will get to stable state.

Question 6 required linearising the system created in SimMechanics using the `linmod` function in the following way:

```
[A,B,C,D]=linmod('coursework_model_template')
```

This method allowed to create the linearised state-space model of the SimMechanics model, which could be investigated with eigenvalues for its stability. Question 6 focused on change in stability of the model when the stiffness of the king-pin mounting, k, is varied from 5000 to 100000 N/m, with other parameters fixed at their default values (speed of the body equal to 10, relaxation length σ fixed at 0.2, and the steer angle δ at zero degrees). With utilisation of for loop and `eig` function, which gives the eigenvalues of the matrix, the MATLAB simulation (Appendix A) gave the following eigenvalue plot.

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Number: 1 Author: Simos Evangelou Date: Indeterminate
Simos Evangelou: What about C?

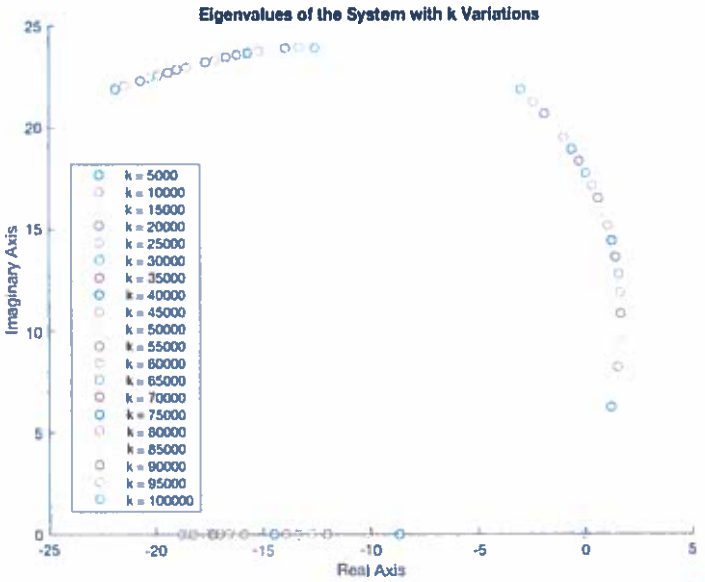


Figure 3. Eigenvalues of the System with k Variations

It is possible to see that each stiffness, k, of the king-pin mounting has given three different eigenvalues on the positive half complex plane (since negative half plane is in symmetry).

For the first set, at the top, as the k value increases the eigenvalues increase in real and imaginary axis. This means it takes more oscillations to get stable, but the frequency of the oscillations increases. Thus, the eigenvalues in the first set shows similar stability. For the second set, at the right, the eigenvalues show that the systems with k value from 5000 to 70000 are unstable with negative real part of the eigenvalues. The third set, at the bottom, portrays as the stiffness k increases, the system gets to the stability state faster. Lastly, all the systems with k variation have an eigenvalue at the origin, which means they have steady state values.

Combining all the eigenvalue information, it is possible to conclude that, the stiffness k of the king-pin mounting should be higher than 70000 to have a stable system or 100000 for the best. This is because, despite other eigenvalues, the ones closest to the imaginary axis dominates the characteristic of the system.

Rotational and Lateral Displacement with k Variations

For the wheel shimmy, the stability of the system will be shown through rotational and lateral displacement. Therefore, Question 7, required the inspection of rotational and lateral displacement with stiffness of king-pin mounting, $k = 30000$ and $k = 100000$, and check whether the result matches with the data collected from the eigenvalue plot. The simulation took place under similar initial conditions as in Question 6 (speed of the body equal to 10, relaxation length σ fixed at 0.2, and the steer angle δ at 0.01 rad),

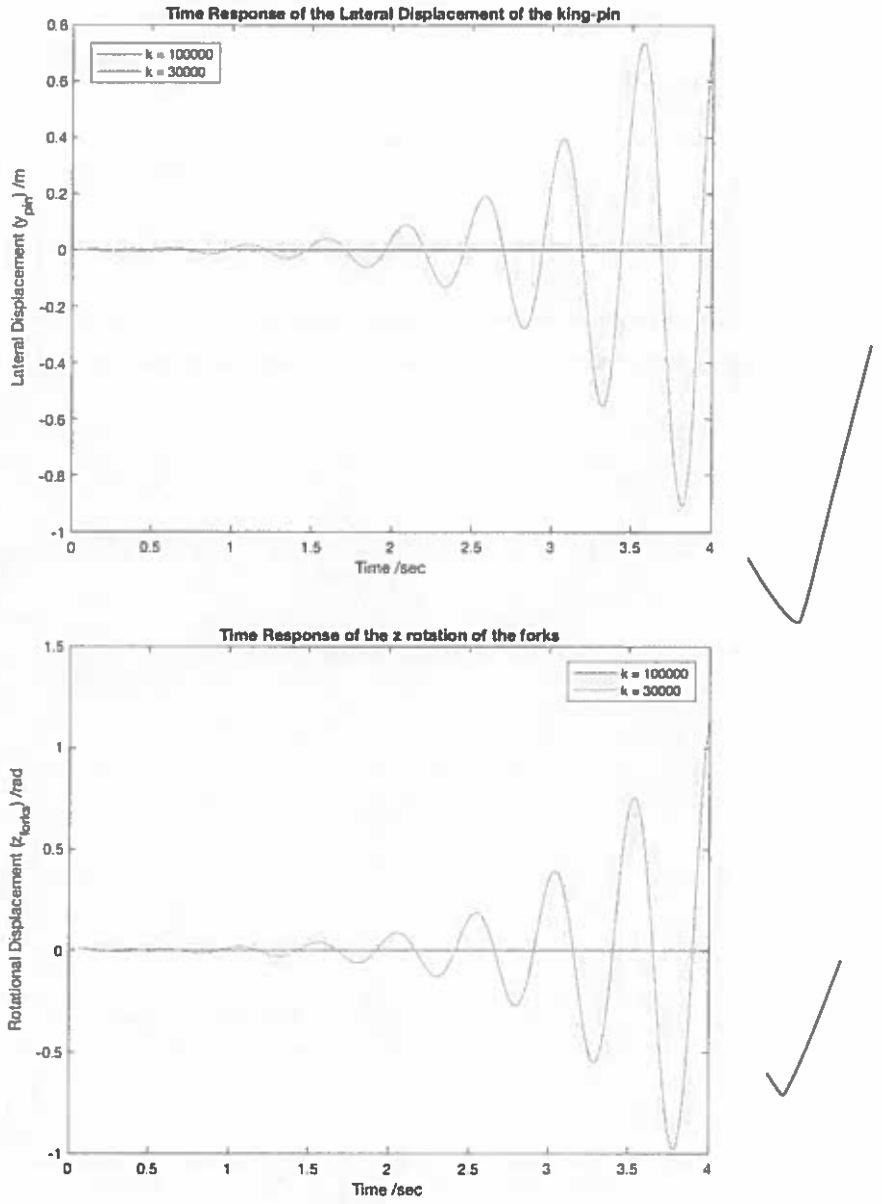


Figure 4. Time Respone of the Lateral and Rotational Displacements with k = 30000 and 100000

Comparing the rotational and lateral displacements of the systems with $k = 30000$ and $k = 100000$ it can be clearly seen that system with $k = 30000$ is highly more unstable than $k = 100000$. The size of oscillation is conspicuously larger, and the time response of the system with $k = 30000$ is getting to infinity, which means the system itself is unstable. The time response of the system with $k = 100000$ is shown as a straight line, which means the system itself is stable. By zooming in, it is possible to see that the system with $k = 100000$ also has oscillations, but it is a stable system with significantly smaller oscillation size compared to when $k = 30000$, so that it is not visible when both graphs are plotted on the same plane.

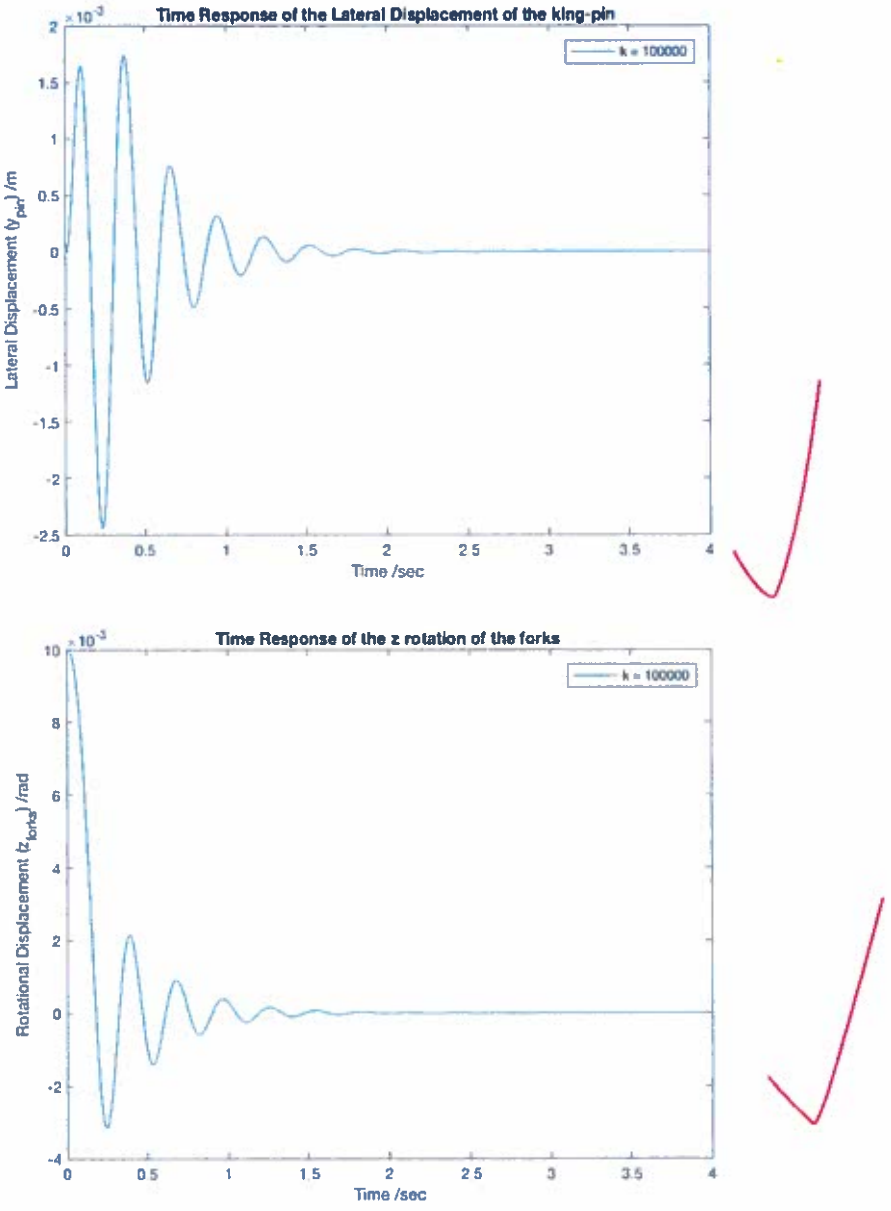


Figure 5. Time Response of Lateral and Rotational Displacements with k = 100000

These results matches with what has been seen in complex plane eigenvalue plot in Question 6. Looking at the second eigenvalues set on the right of the eigenvalue plot, the eigenvalue of the system with $k = 30000$ is on the right hand side of the complex plane, meaning that the system is unstable. However, the eigenvalue of the system with $k = 100000$ is on the negative side of the complex plane, meaning that the system is stable. Also from all the three eigenvalues sets, except for the ones at the origin, show that system with $k = 100000$ is significantly more stable than $k = 30000$, because the eigenvalues of the system with $k = 100000$ stays further away from the imaginary axis then those of the system with $k = 30000$.

Complex Plane Eigenvalues of the System with Speed Variations

With knowledge of relationship between eigenvalue of the systems and the stability of the system, other variables have been varied to see their effects on the stability of the system. Question 8 required a sweep in speed of the body to see its effect on the stability of the system.

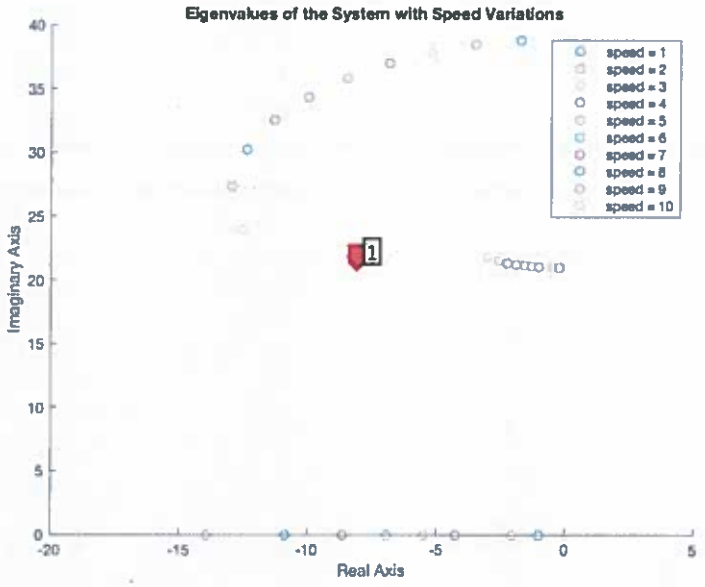


Figure 6. Eigenvalues of the System with Speed Variations

The initial values set for this simulation were relaxation length σ fixed at 0.2, the steer angle δ at zero radians, and the stiffness k of king-pin mounting equal to 100000. Then, the eigenvalue plot portrayed that with $k = 100000$, the system is always stable with its speed variations. But, the eigenvalue plot also shows highlights that the increase in speed gives more stability to the system. This is because the dominating eigenvalues set on the rightmost of the graph has eigenvalues getting further and further away from the imaginary axis as the speed increases.

This result can also be understood intuitively, because the higher the speed of the moving assembly, the lesser the chances of the wheel moving in random motion giving less instability to the body.

Effect of Relaxation Length to Stability of the System

Lastly, the effect of relaxation length σ to the stability of the system has been investigated through eigenvalue plot and time response in lateral and rotational displacement. The initial condition was $speed = 10$, stiffness k of king-pin mounting $= 10000$, and steer angle $\delta = 0$ for eigenvalue plot, and the same but $\delta = 0.01$ and $\sigma = 1$ for the time response.

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Number: 1 Author: Simos Evangelou Date: Indeterminate

Simos Evangelou: Why did you only show up to speed=10m/s and not 50m/s as asked? The part that you showed is correct anyway.

Number: 2 Author: Simos Evangelou Date: Indeterminate

Simos Evangelou: ?

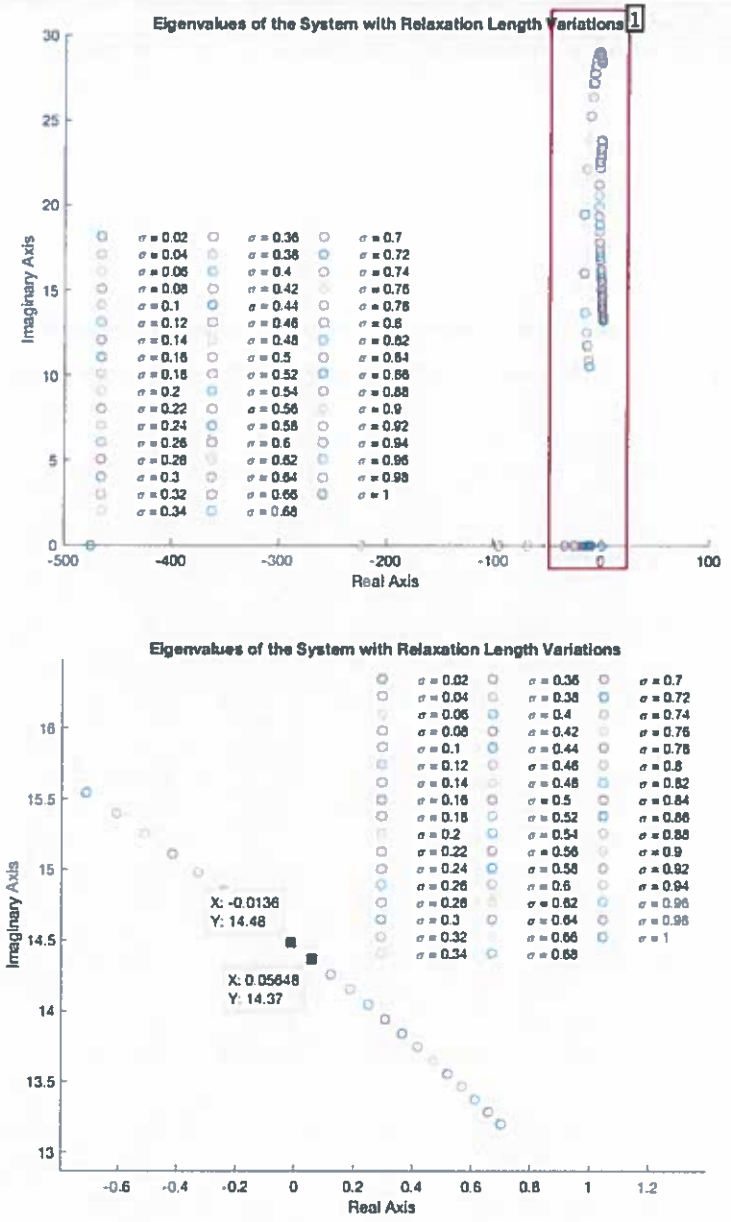


Figure 7. Eigenvalues of the System with Relaxation Length Variations

Unlike other simulations run, the increase in relaxation length σ made the system less stable. As the relaxation length increased the eigenvalues got further away from the imaginary axis and then eigenvalues got closer to imaginary axis again. Then after $\sigma = 0.74$, eigenvalues passed to the right-hand side of the complex plane, meaning the system is unstable. The most stable point from the dominant eigenvalues set at the rightmost of the plane seems to be when $\sigma = 0.24$ where the x-axis value is -3.4.

Page: 10

Number: 1 Author: Simos Evangelou Date: Indeterminate

Simos Evangelou: Yuo could have zoomed here to show more clearly the eigenvalues close to the imaginary axis. Nevertheless, it still looks correct.

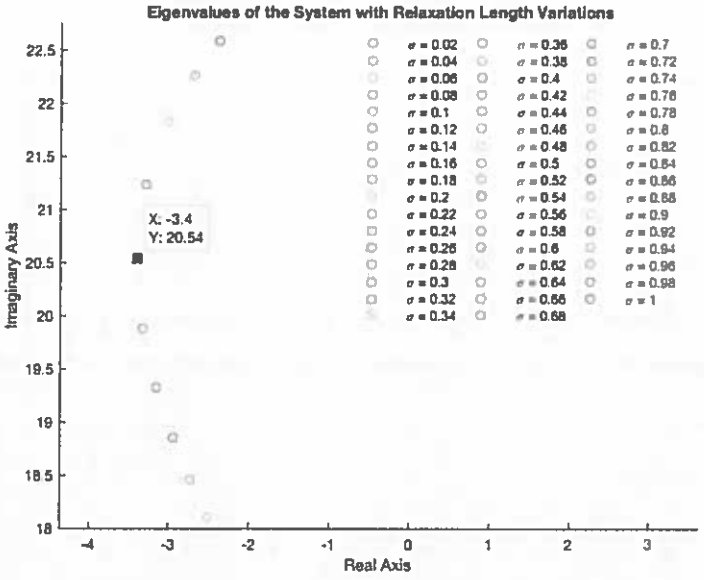


Figure 8. Eigenvalues of the System with Relaxation Length Variation (Zoom-in for Stable Point Inspection)

The time response of the lateral displacement of the king-pin and the z rotation of the forks when $\sigma = 1$ matches with the eigenvalue plot. When $\sigma = 1$, the eigenvalue of the system is at the right-hand side of the complex plane with x-value equal to 0.7001. Having eigenvalue on the right-hand side of the complex plane means the system is unstable and it is shown accurately on both time responses by having the oscillation size increasing towards infinity.

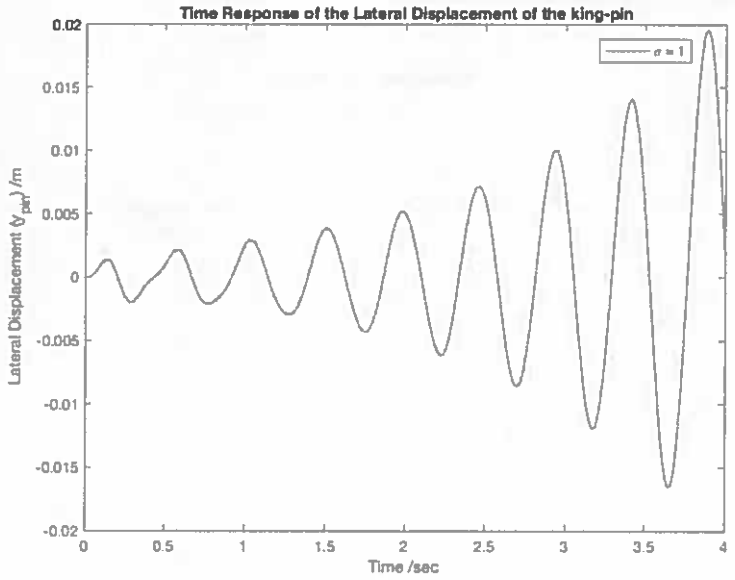


Figure 9. Time Response of the Lateral Displacement of the king-pin (σ = 1)

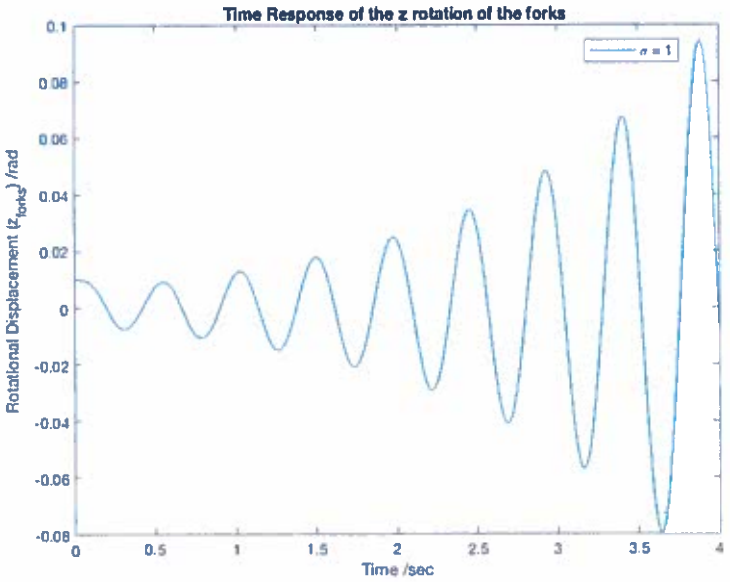


Figure 10. Time Response of the z rotation of the forks (σ = 1)

Controlling the System

Inspecting Eigenvalues of the System

The final task involves controlling the system to stabilise an unstable wheel shimmy system. The model is with forward speed of 10 m/s, relaxation length $\sigma = 0.2$ and stiffness of the king-pin mounting $k = 60000\ N/m$.

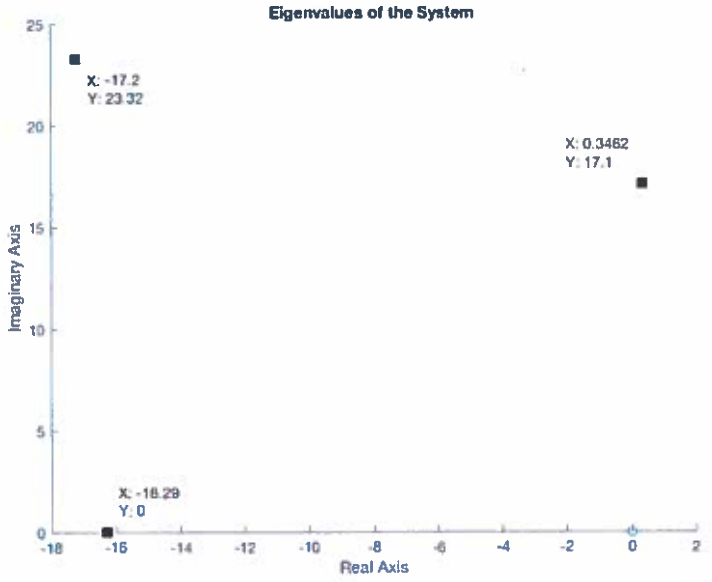


Figure 11. Eigenvalues of the System (Before Controlled)

Computing the eigenvalues of the linearized system, there are three eigenvalues. They are places at $(-17.2, 23.32)$, $(0.3462, 17.1)$, $(-16.29, 0)$, two of which $(-17.2, 23.32)$ and $(-16.29, 0)$ are stable poles, and one of which $(0.3462, 17.1)$ is an unstable pole. The most dominant eigenvalue is the one closest to the imaginary axis which is the one at $(0.3462, 17.1)$. Therefore, despite the two clearly stable eigenvalue points, since the dominant eigenvalue is at the right-hand side of the complex plane, the system is unstable. This could be clearly seen when the time simulation was run for lateral displacement and z rotation of the forks.

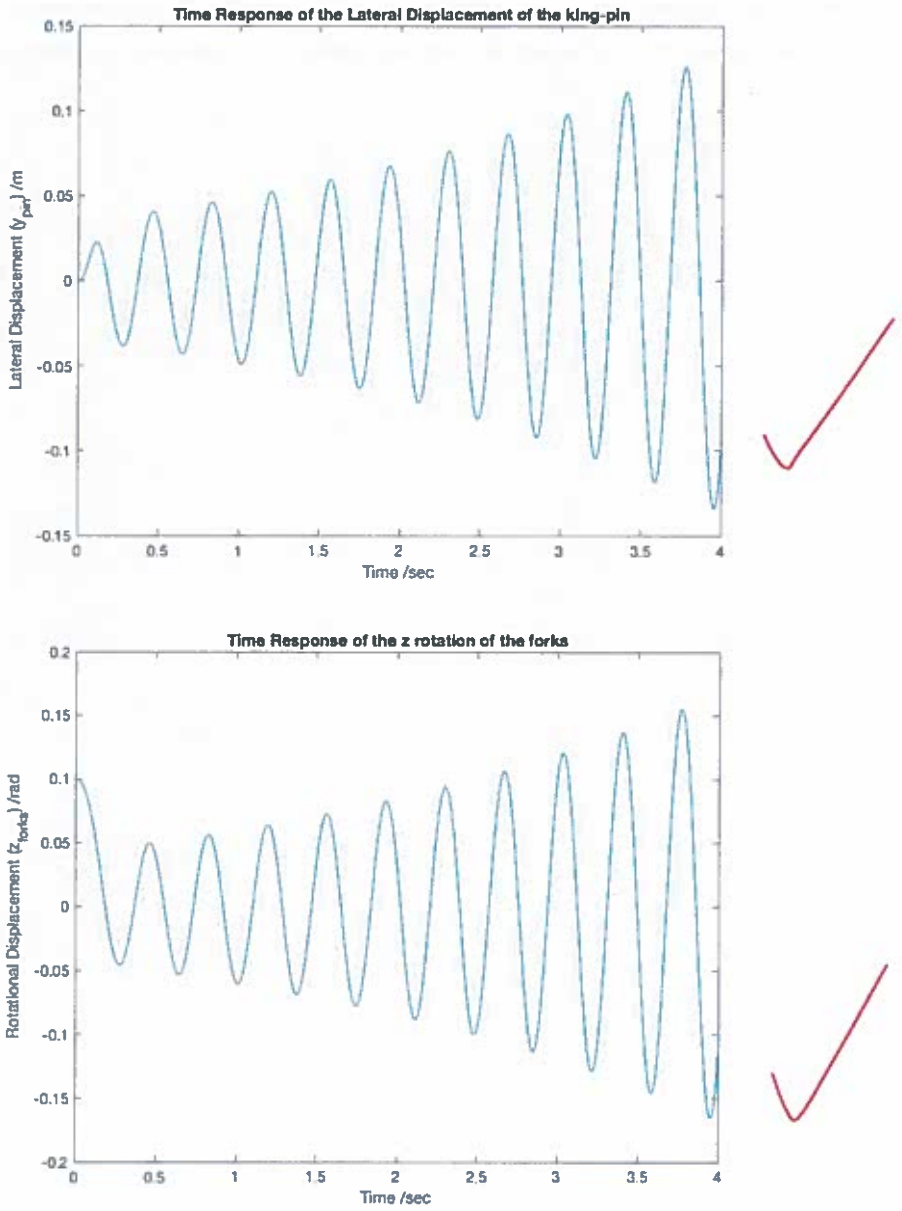


Figure 12. Time Response of Lateral and Rotational Displacements (Before Controlled)

Number: 1 Author: Simos Evangelou Date: Indeterminate
Simos Evangelou: So how many unstable poles are there overall?

Since the dominant eigenvalue is an unstable pole, both time simulations show increase in the size of the oscillation towards infinity.

Feedback Control Law

In order to stabilize the system, the coursework required implementation of a feedback control law

$$T = -\lambda\dot{\delta},$$

where T is a torque applied on the wheel assembly about the steer axis and $\dot{\delta}$ is the steering velocity.

Nyquist Plot for Stability Check

The key to make the best feedback was to find the best λ value. In order to find the best λ, current system with the feedback had to be analyzed. Therefore, a Nyquist plot with an open-loop feedback had been plotted.

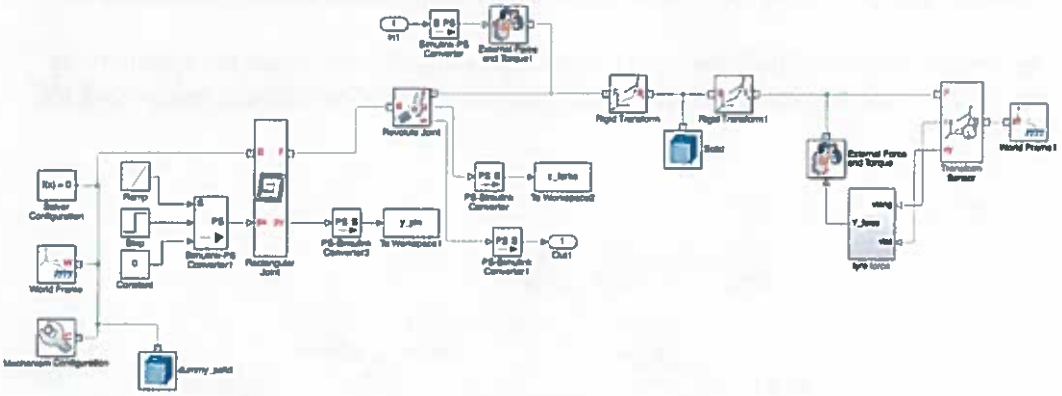


Figure 13. SimMechanics Model of Wheel Shimmy (Open-loop for Nyquist)

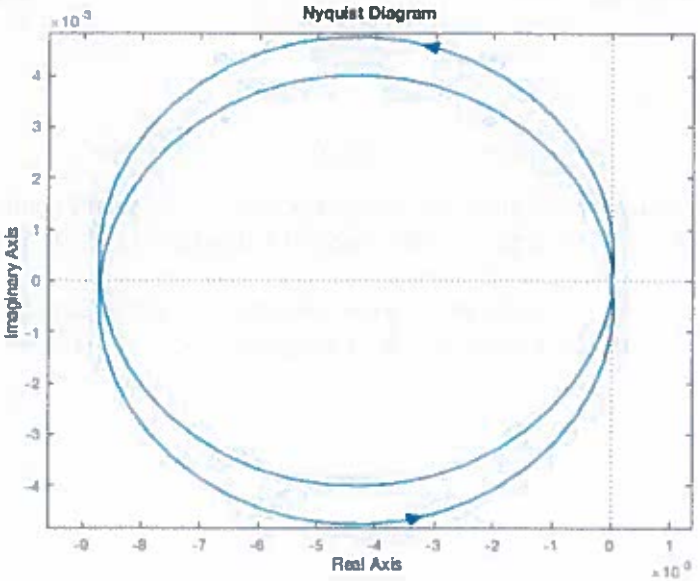


Figure 14. Nyquist Diagram of the System

By adding another dimension of speed to the *Resolute Joint*, which represents the steering, in the settings menu, the form of output for $\dot{\delta}$ was able to be achieved. Then through the addition of *External Force and Torque 1* component, with torque in z direction which represents the direction along the steer axis, the input side of the open-loop could be simulated.

The Nyquist stability criterion states

$$Z = N + P,$$

where Z is the number of unstable closed loop pole, N is the number of clockwise encirclement around $-1 + 0j$ point, and P is the number of unstable open-loop pole.

Looking at this plot, we cannot see any encirclements around the $-1 + 0j$ point, but since we had an unstable pole in the eigenvalue plot,

$$Z = 0 + 1 = 1.$$

Therefore, we can assume that currently the closed loop system would still be unstable.

The control question initially required to figure a λ value that can make the system to be marginally stable. A marginally stable system can be achieved by having a phase margin equal to zero.

Closing the Feedback Loop

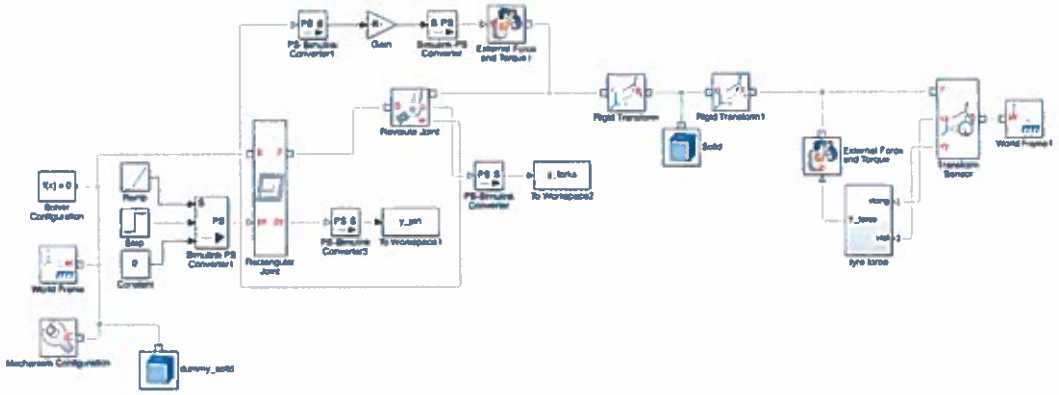


Figure 15. SimMechanics Model of Wheel Shimmy (Closed-loop)

Now closing the loop, the two input and output ports had to be simply connected with the gain, which could be varied by the variable name of `lambda` in MATLAB script.

Now the best value of λ had to be input to complete the closed feedback loop which can stabilize the system. The best value for λ for a marginal stability was 114.6.

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Number: 1 Author: Simos Evangelou Date: Indeterminate

Simos Evangelou: only one?

Number: 2 Author: Simos Evangelou Date: Indeterminate

Simos Evangelou: Why? You did not explain this adequately.

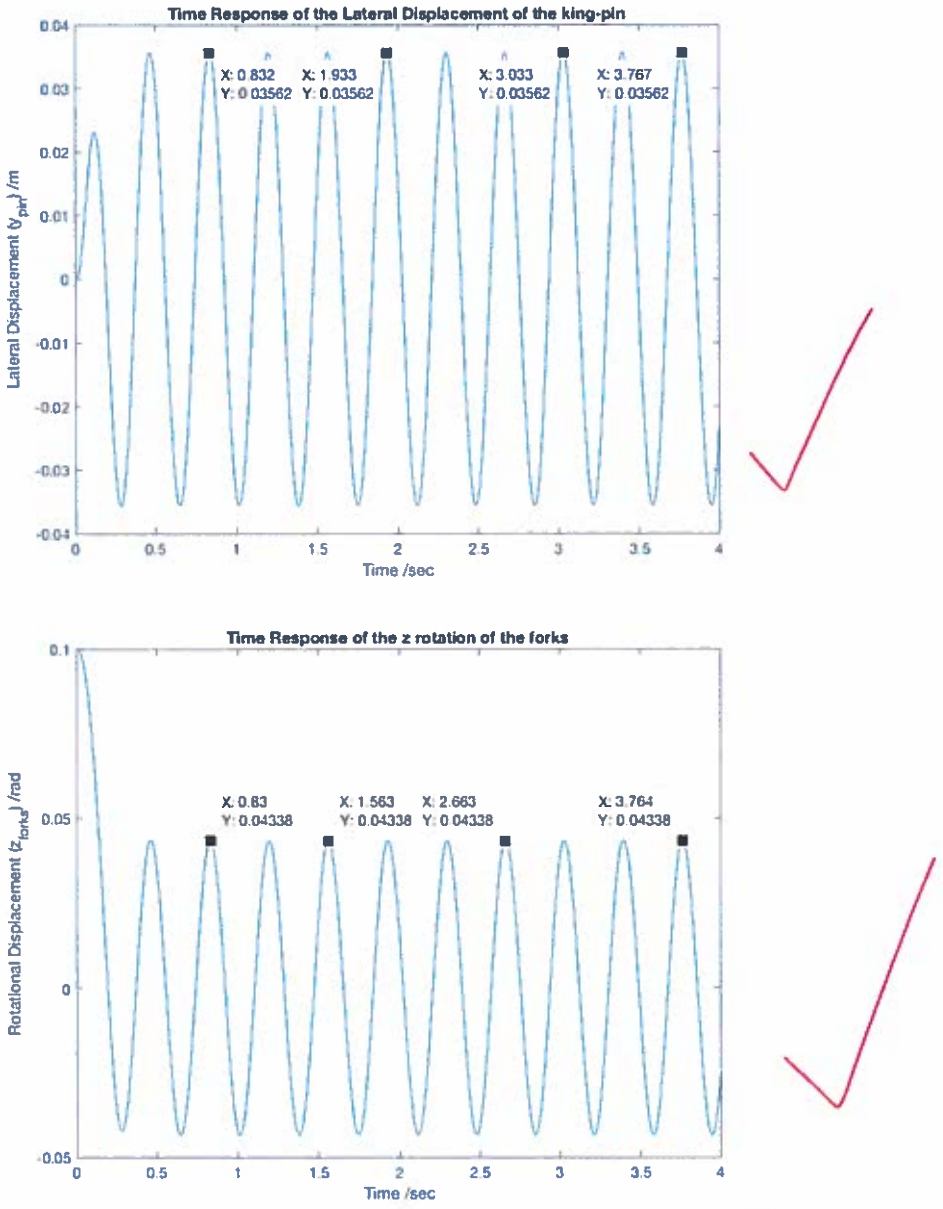


Figure 16. Time Response of the Lateral and Rotational Displacements in Closed-loop ($\lambda = 114.6$)

As viewed on the graphs, the Y values shows the exact same values on any point of the graph after it came to stability. This shows that the system is marginally stable.

In order to make sure that the system is in a stable control through the feedback control law, the double size of λ has also been tested to see if the system is going to stability.

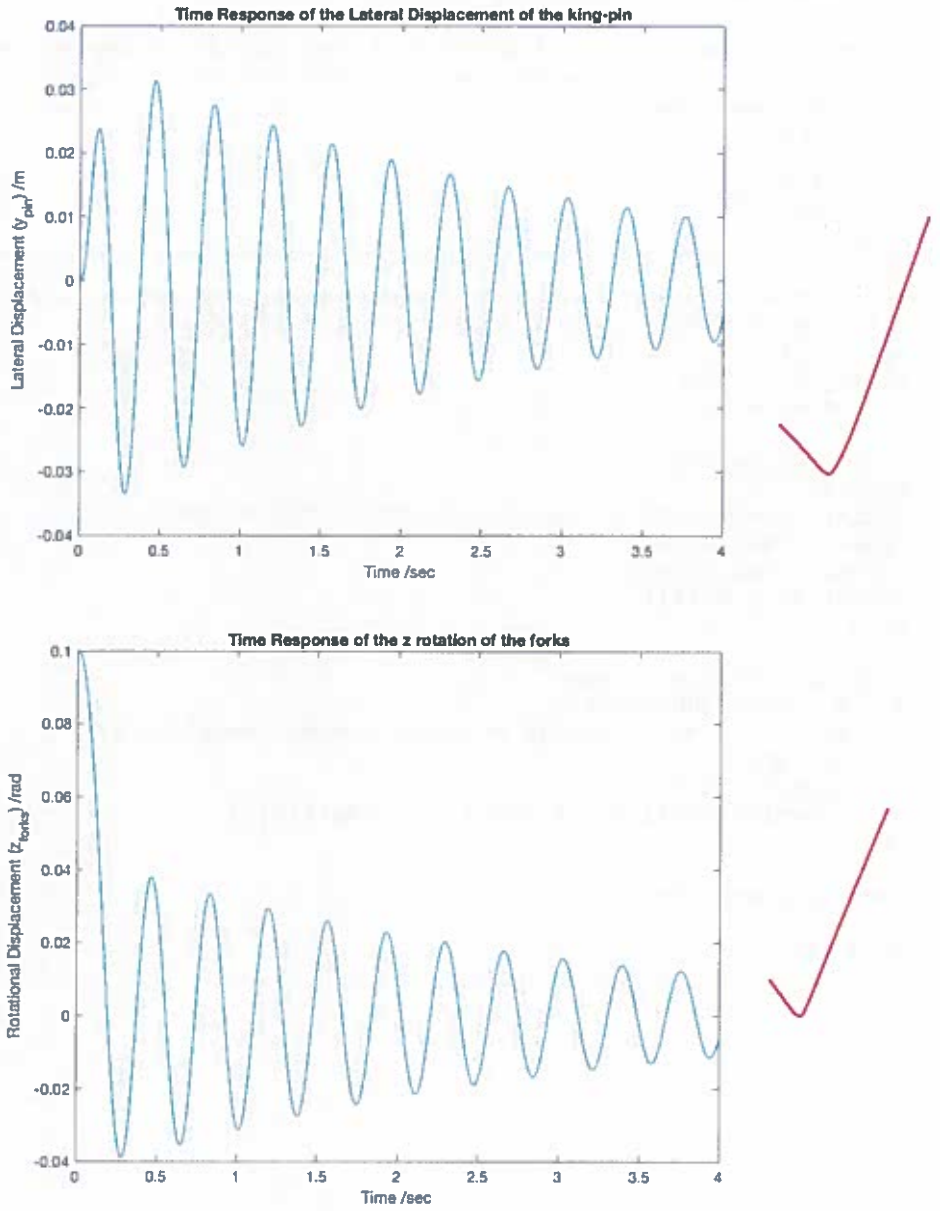


Figure 17. Time Response of the Lateral and Rotational Displacements in Closed-loop ($\lambda = 229.2$)

And as seen on the graph the decreasing size of oscillation has proven that the system is stable. That perturbations could be ameliorated into a stable state.

Appendices

Appendix A: MATLAB script for Question 6

```
% Modelling and Control of Multi-body Mechanical Systems Coursework
% Question 6: Eigenvalue plot with spring-stiffness-sweep
% Author: Seunghyun Lee
% CID: 00843738

% Reset Workspaces
close all;
clear all;
clc;

% Initial Variable Declarations with Initial Values
speed = 10;
sigma_rel = 0.2;
z_initial = 0;

% Eigenvalue Plot Start
figure;
title('Eigenvalues of the System with k Variations')
xlabel('Real Axis')
ylabel('Imaginary Axis')
axis([-25 5 0 25])
hold on;

% Spring Stiffness Sweep
for k = 5000:5000:100000
    [A, B, C, D] = linmod('coursework_model_template');
    e = eig(A);
    plot(e, 'o')
    legendInfo{k/5000} = ['k = ', num2str(k)];
end

legend(legendInfo);

hold off;
% Eigenvalue Plot End
```

Appendix B: MATLAB script for Question 7

```
% Modelling and Control of Multi-body Mechanical Systems Coursework
% Question 7: Time Response of Lateral and Rotational Displacement
% with k Variations
% Author: Seunghyun Lee
% CID: 00843738

% Reset Workspaces
close all;
clear all;
clc;

% Initial Variable Declarations with Initial Values
speed = 10;
sigma_rel = 0.2;
z_initial = 0.01;

% Simulation 1
k = 30000;
sim('coursework_model_template')

sim1_y_time = y_pin.time;
sim1_y_values = y_pin.signals.values;
sim1_z_time = z_forks.time;
sim1_z_values = z_forks.signals.values;

% Simulation 2
k = 100000;
sim('coursework_model_template')

% Graph Plots
figure;
plot(y_pin.time, y_pin.signals.values, sim1_y_time, sim1_y_values)
title('Time Response of the Lateral Displacement of the king-pin')
xlabel('Time /sec')
ylabel('Lateral Displacement (y_{pin}) /m')
legend('k = 100000', 'k = 30000')

figure;
plot(z_forks.time, z_forks.signals.values, sim1_z_time, sim1_z_values)
title('Time Response of the z rotation of the forks')
xlabel('Time /sec')
ylabel('Rotational Displacement (z_{forks}) /rad')
legend('k = 100000', 'k = 30000')

figure;
plot(y_pin.time, y_pin.signals.values)
title('Time Response of the Lateral Displacement of the king-pin')
xlabel('Time /sec')
ylabel('Lateral Displacement (y_{pin}) /m')
legend('k = 100000')

figure;
plot(z_forks.time, z_forks.signals.values)
title('Time Response of the z rotation of the forks')
xlabel('Time /sec')
ylabel('Rotational Displacement (z_{forks}) /rad')
legend('k = 100000')
```

Appendix C: MATLAB Script for Question 8

```
% Modelling and Control of Multi-body Mechanical Systems Coursework
% Question 8: Eigenvalue plot with speed-sweep
% Author: Seunghyun Lee
% CID: 00843738

% Reset Workspaces
close all;
clear all;
clc;

% Initial Variable Declarations with Initial Values
sigma_rel = 0.2;
z_initial = 0;
k = 100000;

% Eigenvalue Plot Start
figure;
title('Eigenvalues of the System with Speed Variations')
xlabel('Real Axis')
ylabel('Imaginary Axis')
axis([-20 5 0 40])
hold on;

% Speed Sweep
for speed = 1:10
    [A, B, C, D] = linmod('coursework_model_template');
    e = eig(A);
    plot(e, 'o')
    legendInfo{speed} = ['speed = ', num2str(speed)];
end

legend(legendInfo);

hold off;
% Eigenvalue Plot End
```

Appendix D: MATLAB Script for Question 9

```
% Modelling and Control of Multi-body Mechanical Systems Coursework
% Question 9: Eigenvalue and Stability plot w/ relaxation-length-sweep
% Author: Seunghyun Lee
% CID: 00843738

% Reset Workspaces
close all;
clear all;
clc;

% Initial Variable Declarations with Initial Values
speed = 10;
z_initial = 0;
k = 100000;

% Eigenvalue Plot Start
figure;
title('Eigenvalues of the System with Relaxation Length Variations')
xlabel('Real Axis')
ylabel('Imaginary Axis')
axis([-500 100 0 30])
hold on;

% Relation Length Sweep
for sigma_rel = 0.02:0.02:1
    [A, B, C, D] = linmod('coursework_model_template');
    e = eig(A);
    plot(e, 'o')
    index = int64(sigma_rel / 0.02);
    legendInfo{index} = ['\sigma = ', num2str(sigma_rel)];
end

columnlegend(3,legendInfo);
% http://uk.mathworks.com/matlabcentral/fileexchange/27389-simonhenin-
columnlegend?
hold off;
% Eigenvalue Plot End

% Stability Check with Time Response of Lateral and Rotational
Displacement
z_initial = 0.01;
sigma_rel = 1;
sim('coursework_model_template')

figure;
plot(y_pin.time, y_pin.signals.values)
title('Time Response of the Lateral Displacement of the king-pin')
xlabel('Time /sec')
ylabel('Lateral Displacement (y_{pin}) /m')
legend('\sigma = 1')

figure;
plot(z_forks.time, z_forks.signals.values)
title('Time Response of the z rotation of the forks')
xlabel('Time /sec')
ylabel('Rotational Displacement (z_{forks}) /rad')
legend('\sigma = 1')
```

Appendix E: MATLAB Script for Question 10 (Eigenvalue plot & Displacements)

```
% Modelling and Control of Multi-body Mechanical Systems Coursework
% Question 10: Eigenvalue and Stability plot with Initial Model
% Author: Seunghyun Lee
% CID: 00843738

% Reset Workspaces
close all;
clear all;
clc;

% Initial Variable Declarations with Initial Values
speed = 10;
sigma_rel = 0.2;
z_initial = 0;
k = 60000;

% Eigenvalue Plot
figure;
title('Eigenvalues of the System')
xlabel('Real Axis')
ylabel('Imaginary Axis')
axis([-18 2 0 25])
hold on;
[A, B, C, D] = linmod('coursework_model_template');
e = eig(A);
plot(e, 'o')
hold off;

% Stability Check with Time Response of Lateral and Rotational
Displacement
z_initial = 0.1;

sim('coursework_model_template')

figure;
plot(y_pin.time, y_pin.signals.values)
title('Time Response of the Lateral Displacement of the king-pin')
xlabel('Time /sec')
ylabel('Lateral Displacement (y_{pin}) /m')

figure;
plot(z_forks.time, z_forks.signals.values)
title('Time Response of the z rotation of the forks')
xlabel('Time /sec')
ylabel('Rotational Displacement (z_{forks}) /rad')
```

Appendix F: MATALAB Script for Question 10 (Nyquist Plot)

```
% Modelling and Control of Multi-body Mechanical Systems Coursework
% Question 10: Nyquist Plot with Open-loop Control Model
% Author: Seunghyun Lee
% CID: 00843738

% Reset Workspaces
close all;
clear all;
clc;

% Initial Variable Declarations with Initial Values
speed = 10;
sigma_rel = 0.2;
z_initial = 0;
k = 60000;

% Linearising the Model
[A, B, C, D] = linmod('coursework_model_template');

% Plot Nyquist Diagram
nyquist(A, B, C, D)

figure;
pzmap(A, B, C, D)
```

Appendix G: MATLAB Script for Question 10 (Closed-loop Displacement Plots)

```
% Modelling and Control of Multi-body Mechanical Systems Coursework
% Question 10: Stability plot with Controlled Model
% Author: Seunghyun Lee
% CID: 00843738

% Reset Workspaces
close all;
clear all;
clc;

% Initial Variable Declarations with Initial Values
speed = 10;
sigma_rel = 0.2;
k = 60000;
lambda = -229.2;

% Stability Check with Time Response of Lateral and Rotational
Displacement
z_initial = 0.1;

sim('coursework_model_template')

figure;
plot(y_pin.time, y_pin.signals.values)
title('Time Response of the Lateral Displacement of the king-pin')
xlabel('Time /sec')
ylabel('Lateral Displacement (y_{pin}) /m')

figure;
plot(z_forks.time, z_forks.signals.values)
title('Time Response of the z rotation of the forks')
xlabel('Time /sec')
ylabel('Rotational Displacement (z_{forks}) /rad')
```